

SHARP RANK-ORDER BOUNDS FOR REDUCED RANK-TEN GRAPHS WITH SPARSE NONSINGULAR CORES

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ABSTRACT. The rank-order conjecture predicts that a finite simple graph of real adjacency rank ten, with no isolated vertices and no equal open neighbourhoods, has at most 62 vertices. The unrestricted rank-ten case remains open. We prove the conjectured bound, sharply, whenever the graph contains a connected nonsingular induced subgraph on ten vertices with at most eleven edges. Equivalently, the full-rank core may be a tree, a unicyclic graph, or a bicyclic graph. We generate all 106, 657, and 2678 isomorphism classes in the three core families and find respectively 15, 136, and 719 nonsingular cores. A basic-subgraph factorisation converts extensions of a fixed core into cliques in a finite compatibility graph, and exhaustive integer target searches exclude 53 pairwise compatible noncore profiles in every case. A separately written verifier regenerates each domain, reconstructs determinants, adjugates and compatibility graphs, relabels every search instance, and independently reaches the same conclusion. Certified 62-vertex examples contain nonsingular cores in all three families, proving sharpness. No floating-point rank, spectral or optimization calculation is used.

1. INTRODUCTION

Let $A(G)$ be the adjacency matrix of a finite simple graph G . The graph is *reduced* when it has no isolated vertex and no two vertices have the same open neighbourhood. Equivalently, the rows of $A(G)$ are nonzero and pairwise distinct. For even adjacency rank r , the rank-order conjecture predicts the maximum

$$m(r) = 2^{(r+2)/2} - 2.$$

Thus $m(10) = 62$. The conjecture is known in several lower ranks and special classes, while recent work continues to state the general problem as open [3, 1]. If it is false, a bounded-rank counterexample exists [5].

Our theorem treats the first three complete sparsity layers of a full-rank ten-vertex principal core.

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Theorem 1.1 (Main theorem). *Let G be a reduced graph with $\text{rank}_{\mathbb{R}} A(G) = 10$. Suppose that G contains a connected nonsingular induced subgraph H on ten vertices with at most eleven edges. Then*

$$|V(G)| \leq 62.$$

The bound is attained with H a tree, a unicyclic graph, or a bicyclic graph.

Since a connected ten-vertex graph has at least nine edges, these hypotheses partition exactly into the tree, unicyclic and bicyclic cases. The method is an exact finite reduction, not a numerical spectral experiment. The basic-subgraph reconstruction and compatibility graph go back at least to Ellingham [2]; related fixed-core and star-complement algorithms appear in [7, 8]. The conjectured bound is already known for the narrower class containing an induced $5K_2$ [6]. Results for graphs that are themselves trees or bipartite [4] do not imply Theorem 1.1, whose ambient graph is unrestricted.

Nonsingular unicyclic and bicyclic graphs have an independent spectral literature. The recent paper [9] characterizes unimodular bicyclic graphs and their possible determinants. Our result addresses a different question: all nonsingular ten-vertex cores, including non-unimodular ones, are used as extension bases for the reduced rank-order problem. The unrestricted rank-ten conjecture is not settled here, because a connected rank-ten graph may have only denser nonsingular ten-vertex cores.

2. EXACT RECONSTRUCTION FROM A PRINCIPAL CORE

Lemma 2.1 (Full-rank principal factorisation). *Let $A = A^T \in \mathbb{R}^{n \times n}$ have rank r , and suppose that the principal submatrix $B = A[C, C]$ has order r and is nonsingular. If $X = A[C, V]$, then*

$$(2.1) \quad A = X^T B^{-1} X.$$

Proof. Order the indices in C first and write

$$A = \begin{pmatrix} B & D \\ D^T & E \end{pmatrix}.$$

Block elimination gives a matrix congruent to $\text{diag}(B, E - D^T B^{-1} D)$. Since A and B have the same rank, the Schur complement vanishes. Substitution yields (2.1). $\square$

Let B now be the adjacency matrix of an induced r -vertex core. The *core profile* of a vertex u is $v_u = A[C, u] \in \{0, 1\}^r$, and (2.1) gives

$$(2.2) \quad A_{uv} = v_u^T B^{-1} v_v.$$

Definition 2.2 (Compatibility graph). For a nonsingular zero-diagonal $0/1$ matrix B of order r , define

$$\mathcal{S}_B = \left\{ v \in \{0, 1\}^r \setminus \{0\} : v^T B^{-1} v = 0 \right\} \setminus \{Be_1, \dots, Be_r\}.$$

The compatibility graph $\Gamma(B)$ has vertex set $\mathcal{S}_B$, with distinct v, w adjacent precisely when

$$(2.3) \quad v^\top B^{-1} w \in \{0, 1\}.$$

Proposition 2.3 (Lossless fixed-core reduction). *Let G be reduced of rank r , and let B be a nonsingular induced r -vertex core. The profiles outside the core form a clique of size $|V(G)| - r$ in $\Gamma(B)$. Conversely, any clique in $\Gamma(B)$, together with the r columns Be_i , reconstructs a reduced simple graph of rank r containing B .*

Proof. The zero diagonal makes every profile isotropic. A zero profile gives an isolated vertex by (2.2). Equal profiles give equal full rows and hence equal open neighbourhoods. An outside profile equal to Be_i duplicates the row of core vertex i . Reducedness excludes all three possibilities, and (2.2) gives (2.3).

Conversely, (2.2) assigns every off-diagonal entry the value 0 or 1, while isotropy gives a zero diagonal. Also

$$(Be_i)^\top B^{-1} v = e_i^\top v = v_i \in \{0, 1\},$$

so every core column is compatible with every binary profile. Distinct nonzero profiles give distinct nonzero rows on the core coordinates. The result is reduced. Equation (2.1) proves rank at most r , and the principal block B proves rank at least r . $\square$

At $r = 10$, an order-63 graph would force a clique of 53 vertices in $\Gamma(B)$. Theorem 1.1 reduces exactly to excluding that target for every nonsingular sparse core B .

3. COMPLETE SPARSE-CORE DOMAINS

3.1. Trees. Canonical leaf extension generates all unlabelled trees: every nontrivial tree has a leaf, whose deletion gives a tree of the preceding order. A centre-rooted AHU word removes duplicates. The free-tree counts at orders $1, \dots, 10$ are

$$(3.1) \quad 1, 1, 1, 2, 3, 6, 11, 23, 47, 106.$$

Rooting every independently generated free tree gives the rooted counts

$$(3.2) \quad 1, 1, 2, 4, 9, 20, 48, 115, 286, 719.$$

Exactly 15 ten-vertex trees are nonsingular. Each has a unique perfect matching and determinant $(-1)^5 = -1$.

3.2. Unicyclic graphs. A connected unicyclic graph has a unique cycle. Deleting the cycle edges leaves a rooted tree at each cycle vertex. Hence an unlabelled unicyclic graph is represented by a cyclic list of rooted-tree words, modulo rotation and reflection. This gives 657 ten-vertex classes. Independently, adding one nonedge to every free ten-vertex tree gives the

same isomorphism set. Exact determinants split the domain into 521 singular and 136 nonsingular cores.

3.3. Bicyclic graphs. Every connected bicyclic graph is obtained both by adding one nonedge to a connected unicyclic graph and by adding two nonedges to a spanning tree. For a structural canonical form, repeatedly delete leaves. The resulting two-core has minimum degree two and cyclomatic number two, so

$$(3.3) \quad \sum_v (\deg(v) - 2) = 2.$$

It has either one vertex of degree four or two vertices of degree three. The former is a figure-eight core. In the latter case, the core is a theta when it has no bridge and a dumbbell otherwise. Rooted attachment words, path permutations and reversals give a complete canonical code.

The unicyclic-plus-nonedge and tree-plus-two-nonedges routes agree on 2678 classes:

$$(3.4) \quad 345 \text{ dumbbells}, \quad 514 \text{ figure-eights}, \quad 1819 \text{ theta types}.$$

Exact determinants give 1959 singular and 719 nonsingular cores, with distribution

$$(3.5) \quad \begin{array}{c|cccccc} \det B & -9 & -5 & -4 & -1 & 0 & 3 & 4 \\ \hline \text{count} & 3 & 6 & 62 & 625 & 1959 & 17 & 6. \end{array}$$

The determinant characterization in [9] is consistent with, but not used as a premise of, this finite enumeration.

4. INTEGER COMPATIBILITY CERTIFICATES

For every nonsingular core, the programs use

$$B^{-1} = \frac{\text{adj}(B)}{\det B}.$$

Thus isotropy and compatibility are tested by integer conditions

$$(4.1) \quad v^\top \text{adj}(B)v = 0, \quad v^\top \text{adj}(B)w \in \{0, \det B\}.$$

All $2^{10} - 1 = 1023$ nonzero binary profiles are inspected; the ten core columns are identified, verified universal, and removed.

At a decision-tree node, let K be the current clique and P the remaining candidates compatible with every member of K . A greedy proper colouring of $\Gamma(B)[P]$ into q independent classes gives the rigorous bound

$$(4.2) \quad \omega(\Gamma(B)[P]) \leq q.$$

If $|K| + q < 53$, the branch is impossible. Otherwise the least remaining profile is included or excluded recursively. This is an exhaustive target decision; a greedy clique is used only as a lower witness and never as a negative certificate.

No target-53 clique exists in any nonsingular instance. Proposition 2.3 proves the upper bound in Theorem 1.1.

TABLE 1. Complete exact target-53 exclusions.

core family	classes	nonsingular	singular	source nodes	audit calls
trees	106	15	91	882,626	760,220
unicyclic	657	136	521	9,987,213	11,600,208
bicyclic	2678	719	1959	48,967,729	41,704,342

5. INDEPENDENT AUDIT AND FAIL-CLOSED DESIGN

The decisive computations were repeated by implementations that import no discovery module and do not trust discovery canonical labels. For trees, the verifier changes representatives and regenerates the free-tree domain. For unicyclic graphs it independently decorates cycles with bracket-coded rooted trees. For bicyclic graphs it starts from the independently generated free trees, adds two nonedges in all 66,780 labelled cases, and uses a separately written bracket-word two-core code. The resulting edge-list sets agree with the discovery sets as isomorphism sets.

For every nonsingular core, the audit computes the adjugate from cofactors and checks both identities

$$(5.1) \quad B \operatorname{adj}(B) = \operatorname{adj}(B)B = (\det B)I.$$

It reconstructs all profiles and compatibility edges from (4.1), then cyclically relabels every target instance. Before the large searches, its target engine is compared with literal subset enumeration on every labelled graph through order five and on 100 deterministic random graphs at each order six through nine, for 1500 test graphs.

The bicyclic certificate undergoes five fail-closed attacks: wrong source hash, a dropped record, an edge list with a self-loop, disabled exact search, and a flipped top-level conclusion. All are rejected. No floating-point eigenvalue, rank, optimization value or heuristic stopping rule enters the proof.

The supplementary manifest binds the source, independent verifier, theorem, sharpness witness, attacks and audit reports. Its SHA-256 digest is

da941f91a710dccf713c0c113350e65ac7faa4405bef65afbd8dd6d7dc1f8f40.

6. SHARPNESS

The standard iterated construction yields a reduced graph G_{62} of order 62 and exact adjacency rank ten. Its certificate checks symmetry, binary entries, zero diagonal, no isolates, pairwise distinct open neighbourhoods, a nonzero 10×10 principal determinant, and all $62^2 = 3844$ entries of the rank-ten factorisation.

The same graph contains the following three nonsingular induced cores; the indices are zero-based ambient vertex labels in the supplementary adjacency

matrix.

	family	ambient indices	$ E(H) $	$\det A(H)$
(6.1)	tree	0, 1, 2, 4, 6, 12, 14, 28, 30, 60	9	−1
	unicyclic	1, 2, 4, 6, 12, 14, 28, 30, 60, 10	10	−1
	bicyclic	2, 4, 6, 12, 14, 28, 30, 60, 10, 7	11	−1.

Connectivity and the edge counts make the induced cores respectively acyclic, unicyclic and bicyclic. Thus G_{62} attains the bound in every part of Theorem 1.1.

7. SCOPE AND REPRODUCIBILITY

The theorem is conditional on the presence of a connected nonsingular ten-vertex induced core with at most eleven edges. It does not assert that every connected rank-ten graph has such a core. The next unclosed finite layer consists of connected nonsingular ten-vertex cores with twelve edges (cyclomatic number three). Hence the unrestricted rank-ten rank-order conjecture remains open.

The supplementary archive contains the complete labelled core domains, integer source certificates, independent audit outputs, fail-closed attack record, sharpness witnesses, theorem report and manifest. A clean replay requires only Python’s standard library. The theorem can therefore be checked without trusting floating-point libraries, graph databases, or the authors’ machine state.

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REFERENCES

1. Saieed Akbari, Clive Elphick, Hitesh Kumar, Shivaramakrishna Pragada, and Quanyu Tang, *A new conjecture on the inertia of graphs*, Discrete Mathematics **349** (2026), no. 4, 114953.
2. Mark N. Ellingham, *Basic subgraphs and graph spectra*, Australasian Journal of Combinatorics **8** (1993), 247–265.
3. H. Esmailian, E. Ghorbani, S. Hossein Ghorban, and G. B. Khosrovshahi, *Maximal graphs with respect to rank*, Discrete Mathematics **344** (2021), no. 1, 112191.
4. Ebrahim Ghorbani, Ali Mohammadian, and Behruz Tayfeh-Rezaie, *Maximum order of trees and bipartite graphs with a given rank*, Discrete Mathematics **312** (2012), no. 24, 3498–3501.
5. ———, *On order and rank of graphs*, Combinatorica **35** (2015), no. 6, 655–668.
6. Willem H. Haemers and M. J. P. Peeters, *The maximum order of adjacency matrices of graphs with a given rank*, Designs, Codes and Cryptography **65** (2012), no. 3, 223–232.
7. Zoran Stanić, *A note on the minimum rank of graphs with given dominating induced subgraph*, American Journal of Combinatorics **3** (2024), 44–53.

8. ———, *Employing star complements in search for graphs with fixed rank*, Linear Algebra and its Applications **718** (2025), 14–29.
9. Md Isheteyak Zaffer, *Unimodular bicyclic graphs*, 2026.

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